

Registration Number:

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DELHI-NCR CAMPUS, GHAZIABAD (U.P.)

SRM Institute of Science and Technology
Department of Computer Science and Engineering
 Delhi – Meerut Road, Sikri Kalan, Ghaziabad, Uttar Pradesh – 201204

Academic Year: 2025-26 (ODD)

Test : Internal Examination I

Course Code & Title : 21CSE415T& Fuzzy logic and its applications

Year & Sem : 4th & 7th

Date & Session : 02/09/25&AN

Duration: 1 Hour 30 Minutes

Max. Marks: 50

Part - A

Answer all questions

(10 x 1 = 10 Marks)

Q. No	Question	Marks	BL	CO	PO
1	The composition of two fuzzy relations is performed using a) Intersection and union b) Min-max or max-min operations c) Addition and subtraction d) Differentiation	1	2	1	2
2	Which operation in fuzzy sets corresponds to the intersection of classical sets? a) Maximum b) Minimum c) Addition d) Subtraction	1	1	1	1
3	Specify the role that membership functions do play in fuzzification? a) They convert fuzzy output to crisp output b) They assign degrees of membership to crisp inputs c) They generate fuzzy rules d) They perform defuzzification	1	1	1	1
4	The fuzzy equivalence relation is also known as a) Fuzzy congruence b) Fuzzy similarity relation c) Fuzzy implication d) Fuzzy complement	1	1	1	1
5	The height of a fuzzy set is defined as: a) The total number of elements in the set b) The average membership value c) The maximum membership value among all elements d) The difference between the highest and lowest membership values	1	2	1	2
6	Which matrix in the RLS algorithm helps in weighting the importance of new versus old data? a) Covariance matrix b) Identity matrix c) Hessian matrix d) Transition matrix	1	3	2	3
7	In decision-making, rank ordering helps to a) Randomly shuffle options b) Eliminate the need for criteria c) Prioritize alternatives based on preferences or performance d) Guarantee optimal results	1	2	2	2

8	Fuzzy logic is best suited for which type of problem?	1	1	2	2
	a) Precise mathematical equations				
	b) Problems with binary outcomes only				
	c) Problems involving uncertainty or linguistic variables				
	d) High-speed arithmetic operations				
9	What does the height of a membership function at a specific point represent?	1	2	2	2
	a) The probability of the input value				
	b) The degree of membership of that value in the fuzzy set				
	c) The crisp output value				
	d) The error rate				
10	Identify the potential issue with setting the learning rate too high in gradient descent?	1	3	2	3
	a) The system becomes more accurate				
	b) The algorithm converges faster				
	c) The algorithm may overshoot and fail to converge				
	d) It leads to underfitting				

Part B

Answer any two questions

4M x 2Q = 8 Marks

11	How is the complement of a fuzzy set defined? What is its effect on the membership values?	4	1	1	2
12	Compare and contrast between fuzzy tolerance and equivalence relation.	4	2	1	2
13	Discuss the concept of support and core of a fuzzy set. How do they relate to the uncertainty and definiteness of the fuzzy set?	4	3	1	3

Part C

Answer any two questions

4M x 2Q = 8 Marks

14	Give an example of a scenario where modified learning from examples is applied.	4	3	2	2
15	How would fuzzy logic handle the linguistic term "moderately warm" in a temperature control system?	4	1	2	3
16	Which fuzzy system parameters can be adjusted using the gradient method?	4	2	2	3

Part D

Answer all questions

12M x 2Q = 24 Marks

17	(A) Explain the composition of fuzzy relations. Describe different types of composition methods with suitable examples.	12	2	1	2
	(OR)				
	(B) Describe different types of membership functions used in fuzzification. How do they affect system performance?	12	1	1	2
18	(A) Explain the Batch Least Squares (BLS) algorithm in detail. Discuss its mathematical formulation, working mechanism, advantages, disadvantages, and applications.	12	3	2	3
	(OR)				
	(B) Define inference and explain its role in logical reasoning. What are the types of inference and how do they contribute to knowledge and decision-making?	12	2	2	3

Registration Number:

R	A	2	2	1	1	0	2	6	6	3	0	0	6	4
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Academic Year: 2025-26 (ODD)

Test : Internal Examination I
 Course Code & Title : 21CSE454T & Computer Vision
 Year & Sem : IV & VII

Date & Session : 01.09.2025 (AN)
 Duration: 1 Hour 30 Minutes
 Max. Marks: 50

Part - A

Answer all questions

(1M x 10Q = 10 Marks)

Q. No	Question	Marks	BL	CO	PO
1	Thresholding in image processing is mainly used for a) Image enhancement b) Image segmentation c) Edge detection d) Noise removal	1	L1	1	1
2	Otsu's method of thresholding is based on a) Minimizing edge strength b) Minimizing intra-class variance c) Maximizing histogram entropy d) Median pixel intensity	1	L1	1	1
3	An interest point detector that uses Laplacian of Gaussian (LoG) to detect blobs is called: a) Harris–Stephens detector b) Shi–Tomasi detector c) SIFT d) Hessian–Laplace detector	1	L1	1	1
4	SURF (Speeded-Up Robust Features) is faster than SIFT mainly because it uses a) Haar wavelets and integral images b) Fourier descriptors c) Median filtering d) Thresholding	1	L2	1	1
5	Interest point detection in image processing is mainly used for a) Noise removal b) Shape smoothing c) Feature extraction for matching d) Histogram equalization	1	L1	1	1
6	In image analysis, Zernike moments are used as a) Edge descriptors b) Histogram descriptors c) Region descriptors with rotation invariance d) Noise descriptors	1	L1	2	2
7	The ratio of the area of a region to the area of its bounding box is called: a) Solidity b) Extent c) Compactness d) Circularity	1	L1	2	1
8	Name the region descriptor which is used to measure elongation of an object? a) Eccentricity b) Compactness c) Solidity d) Convexity	1	L1	2	2
9	The Euler number of a region is defined as a) Number of objects + number of holes b) Number of objects – number of holes c) Area × perimeter d) Compactness × eccentricity	1	L2	2	3
10	High-frequency Fourier descriptors mainly capture a) Overall shape b) Noise and fine details of the boundary c) Translation invariance d) Scaling properties	1	L1	2	2

Part B

Answer any two questions

4M x 2Q = 8 Marks

11	Discuss how filters operate in both spatial and frequency domains. Illustrate your answer with the help of diagrams showing their effects on images.	4	L2	1	1
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- 12 Given the following 3x3 image patch:

10	20	50
20	10	50
20	20	50

4 L3 1 1

- 13 Apply Roberts, Prewitt and Sobel operators step-by-step to compute the edge response. Explain Gaussian filtering. Why it is preferred over simple averaging?

4 L2 1 1

Part C

Answer any two questions

4M x 2Q = 8 Marks

- 14 Given an initial contour around an object, explain how the snake evolves towards the boundary.
- 15 Given the following 6x6 binary image, perform connected component analysis on the given image. Show each step clearly.

4 L3 2 3

4 L3 2 2

1	1	0	0	0	1
0	0	1	1	0	0
1	1	1	0	1	0
0	0	1	1	0	1
1	1	1	0	1	0
0	0	1	1	0	1

- 16 In a facial recognition system, parts of the face are occluded by masks. Explain how recognition performance can be enhanced in the given scenario?

4 L4 2 3

Part D

Answer all questions

12M x 2Q = 24 Marks

- 17 (A) Discuss the concept of a corner in an image. With the help of mathematical derivation, explain how the Harris Corner Detector identifies corners.

12 L3 1 1

(OR)

(B) Explain in detail the basic morphological operations with their set-theoretic and algebraic formulations. Explain how the choice of structuring element influences the result of these operations. Discuss their role in shape smoothing and gap filling with suitable examples.

12 L2 1 1

- 18 (A) Consider a simple binary shape (e.g., a rectangle or circle). Sketch how its boundary is represented and describe how Fourier Descriptors capture its essential shape features while ignoring noise and minor variations.

12 L4 2 2

(OR)

(B) With the help of mathematical morphology, derive the expression for skeleton extraction using morphological operations. Show how the skeleton can be reconstructed from its components.

12 L3 2 3

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Academic Year: 2025-26 (ODD)

Test : Internal Examination I
Course Code & Title: 21CSC401J & Deep Learning Techniques
Year & Sem : IV & VII

Date & Session: 03-09-2025 & AN
Duration: 1 Hour 30 Minutes
Max. Marks: 50

Part - A

Answer all questions

(10 x 1 = 10 Marks)

Q. No	Question	Marks	BL	CO	PO
1	The McCulloch-Pitts neuron is primarily used to model: a) Biological action potentials b) Logical functions c) Gradient descent updates d) Activation functions	1	1	1	2
2	The primary purpose of a McCulloch-Pitts neuron? a) Generate action potentials b) Perform logical operations c) Apply gradient descent d) Serve as an activation function	1	1	1	2
3	Why a single perceptron fails to learn XOR. a) Lack of convergence b) Non-linear separability c) Wrong loss function d) High variance	1	1	1	2
4	Sigmoid neuron output for input=0 is: a) 0 b) 0.25 c) 0.5 d) 1	1	2	1	2
5	The perceptron output for AND gate with input (1,0) is. a) 0 b) 1 c) Undefined d) Depends on learning rate	1	2	1	2
6	The vanishing gradient problem is most associated with: a) ReLU b) Sigmoid c) Tanh d) Leaky ReLU	1	1	2	2
7	RMSProp improves AdaGrad by: a) Lower cost b) Handling sparse features c) Avoiding aggressive decay d) Only convex use	1	1	2	2
8	The main purpose of early stopping in training neural networks is: a) Prevent overfitting b) Reduce gradient explosion	1	2	2	2

	c) Increase dataset size				
	d) Improve activation function				
9	The main advantage of using mini-batch gradient descent compared to batch and stochastic gradient descent.	1	1	2	2
	a) Provides a good balance between memory usage and speed of convergence				
	b) Always reaches the global minimum				
	c) Does not require any parameter tuning				
	d) Completely removes randomness in updates				
10	Adam optimizer combines the features of:	1	1	2	2
	a) Momentum & AdaGrad				
	b) RMSProp & Nesterov				
	c) Gradient descent & Mini-batch				
	d) Mini -batch & Momentum				

Part B

Answer any two questions

4M x 2Q = 8 Marks

11	Illustrate the role of bias in a perceptron. Why is it important in learning decision boundaries.	4	3	1	2
12	Outline the backpropagation algorithm for training a neural network.	4	4	1	2
13	Demonstrate the importance of activation functions in neural networks. Why can't we train deep models without them?	4	3	1	2

Part C

Answer any two questions

4M x 2Q = 8 Marks

14	Determine the concept of vanishing and exploding gradients in deep neural networks.	4	3	2	2
15	Illustrate mini-batch gradient descent, batch gradient descent, and stochastic gradient descent.	4	3	2	2
16	Classify the concept of regularization in deep learning.	4	4	2	2

Part D

Answer all questions

12M x 2Q = 24 Marks

17	(A) Describe the gradient descent algorithms. Compare advantages, and limitations with examples	12	4	1	2
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(OR)

	(B) Classify the Backpropagation algorithm in detail. explain how the error is propagated backwards layer by layer to update the weights. Support your answer with a simple two-layer example.	12	4	1	2
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18	(A) Discuss the impact of Hyperparameters on neural network training. Explain learning rate, batch size, momentum, epochs, and weight initialization. How do they affect training accuracy and convergence?	12	4	2	2
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(OR)

	(B) Explain Overfitting in deep networks. Why does it occur, and how can we reduce it using data augmentation, dropout, regularization, and cross-validation?	12	4	2	2
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Academic Year: 2025-26 (ODD)

Test : Internal Examination I
 Course Code & Title : 21ME0112T/Renewable Energy Sources and Application
 Year & Sem : IV/VII

Date & Session: 08/09/25 / AN
 Duration: 1 Hour 30 Minutes
 Max. Marks: 50

Part - A

Answer all questions

(10 x 1 = 10 Marks)

Q. No	Question	Marks	BL	CO	PO
1	Find the name of device which converts solar energy directly into electrical energy. a) Solar collector b) Solar heater c) Photovoltaic cell d) Solar concentrator	1	1	1	1
2	The main advantage of solar energy is a) High operating cost b) Non-renewable in nature c) Eco-friendly and abundant d) Requires fossil fuels	1	1	1	1
3	Which type of solar radiation is most responsible for electricity generation in solar cells? a) Infrared b) Ultraviolet c) Visible light d) X-rays	1	1	1	1
4	The unit of solar radiation intensity is: a) Watt per square meter (W/m^2) b) Joule per second c) Lux d) Candela	1	1	1	1
5	The approximate average solar constant at the Earth's atmosphere is a) $100 W/m^2$ b) $340 W/m^2$ c) $1367 W/m^2$ d) $2000 W/m^2$	1	1	1	1
6	Wind energy is a form of: a) Kinetic energy b) Potential energy c) Thermal energy d) Nuclear energy	1	1	2	7
7	The largest producer of wind energy country (as of 2023) is a) USA b) China c) India d) Germany	1	2	2	7

8	Following is used to measure wind speed	1	2	2	1
	a) Anemometer				
	b) Barometer				
	c) Hygrometer				
	d) Altimeter				
9	The main environmental concern of wind farms is:	1	2	2	1
	a) Carbon emissions				
	b) Noise and impact on birds				
	c) Radioactive waste				
	d) Water pollution				
10	The power available in the wind is proportional to:	1	1	2	1
	a) Wind speed (v)				
	b) v^2				
	c) v^3				
	d) v^4				

Part B

Answer any two questions

4M x 2Q = 8 Marks

11	Explain the attenuation of solar radiation.	4	2	1	1,7
12	Describe the working principles of a solar pond and outline its various applications.	4	2	1	1,7
13	Explain the measurement device of solar radiation.	4	2	1	1,7

Part C

Answer any two questions

4M x 2Q = 8 Marks

14	Describe the term wind energy and origin of wind.	4	2	2	1,7
15	Explain the beam, diffuse and global radiation.	4	2	2	1,7
16	Describe the nature of wind and its application.	4	2	2	1,7

Part D

Answer all questions

12M x 2Q = 24 Marks

17	(A) Illustrate the classification of solar concentrators with neat diagrams.	12	3	1	1,7
	(OR)				
	(B) Describe the working principle of solar photovoltaic system, components and its application.	12	3	1	1,7
18	(A) Illustrate the wind data measurement system.	12	3	2	1,7
	(OR)				
	(B) Describe the wind energy scenario in India and the world.	12	3	2	1,7

Registration Number:	R	A	2	2	1	1	0	2	6	0	3	0	0	6	4
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Academic Year: 2025-26 (ODD)

Test : Internal Examination I Course Code & Title: 21CSC402P & Report Writing Year & Sem : IV Year & 7th Semester	Date & Session: 04.09.25 & AN Duration: 1 Hour 30 Minutes Max. Marks: 50
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Part – A

Answer all questions

(10 x 1 = 10 Marks)

Q. No	Question	Marks	BL	CO	PO
1	Report is often used to display the result of a. Experiment b. Investigation c. Inquiry d. Introduction	1	L1	1	4
2	Which is one of the embodiments of the informal report? a. Feasibility report b. In-Periodic report c. Both A, B d. Infeasibility report	1	L1	1	4
3	In report writing, stating the purpose, scope, and order of sections is termed as a. Title page b. Summary c. Introduction d. Glossary	1	L1	1	4
4	What comes after the objective of report writing process? a. Planning of the report b. Title page of the report c. Summary of the report d. Body of the report	1	L1	1	4
5	In report writing, the concise version that highlights one key idea from each main section is called a. Glossary b. Introduction c. Abstract d. Summary	1	L1	1	4
6	The term 'research methodology' refers to a. the methods used in data collection and analysis b. the rules of writing a research report or paper c. the specific methods of study and analysis d. the theoretical paradigms for data collection, analysis and interpretation	1	L1	2	2
7	The conceptual framework and background of research which will become the source for formulation of the hypothesis is known as a. Methodology b. Review of literature c. Data analysis d. Abstract	1	L1	2	2

8	Which of the following statement is true about the simple random sample? a. Every unit that constitutes the population has an equal chance of being paired with another unit in that given populations as component of the sample. b. Every Unit has an equal chance of being selected as part of the sample. c. Each sample of a fixed size in a given population has equal chance of being selected as the sample. d. Every Unit has an nonequal chance of being selected as part of the sample.	1	L1	2	2
9	The best graphical way to present data related to different levels of a certain variable is: a. Pictogram b. pie chart c. histogram d. frequency polygon	1	L1	2	2
10	Which of these is the best way to establish a proper rapport with audience? a. Pointing a finger b. Making eye contact c. Waving your hands d. Standing erect	1	L1	2	2
Part B					
Answer any two questions		4M x 2Q = 8 Marks			
11	Who are regarded as the audience in report writing? Explain in brief.	4	L2	1	4
12	Identify the key elements included in a feasibility report.	4	L2	1	4
13	Mention the key constituents of a report. Explain in detail.	4	L2	1	4
Part C					
Answer any two questions		4M x 2Q = 8 Marks			
14	List and explain the different types of research methodology.	4	L2	2	2
15	Differentiate between qualitative and quantitative research.	4	L3	2	2
16	What are the ways of communicating and displaying the analyzed data?	4	L2	2	2
Part D					
Answer all questions		12M x 2Q = 24 Marks			
17 (A)	Describe the structure and formatting style used in report writing.	12	L3	1	4
(OR)					
(B)	What are the main components of a report and how is the text of the report structured?	12	L2	1	4
18 (A)	Describe the step-by-step process of effective problem-solving process.	12	L2	2	2
(OR)					
(B)	Outline the guidelines for the preparation and delivery of an oral report writing.	12	L3	2	2